

Functional Annotation Report: *argH* (Argininosuccinate Lyase, P59618) in *Pseudomonas putida* KT2440

Identity Verification (mandatory)

The target is unambiguous and fully consistent across all evidence:

- **Gene / locus:** `argH` / `PP_0184`, *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / CFBP 8728 / NCIMB 11950 / KT2440).
- **Protein:** Argininosuccinate lyase (ASAL; arginosuccinase), EC 4.3.2.1.
- **Family / domains:** lyase 1 family (aspartase/fumarase superfamily); InterPro Arg_succ_lyase_C (IPR029419), Argininosuccinate_lyase (IPR009049), Fumarase/histidase_N (IPR024083), Fumarate_lyase_CS (IPR020557), Fumarate_lyase_fam (IPR000362).

The gene symbol `argH` matches the protein description, and the family/domain assignment is confirmed **directly from the P59618 sequence** (see Key Findings 4 and 5). The `argH` symbol universally denotes argininosuccinate lyase in bacterial genetics, and every retrieved paper concerns the correct enzyme family (argininosuccinate lyase / δ -crystallin / aspartase–fumarase superfamily). **No gene-symbol ambiguity was encountered.** While there is limited literature on the *P. putida* protein specifically, its function is established with high confidence by (i) curated sequence annotation, (ii) near-identical conservation with experimentally characterized orthologs, and (iii) *P. putida*-specific pathway and regulatory studies.

Summary

The gene **argH** (ordered locus **PP_0184**; UniProt **P59618**) of *Pseudomonas putida* KT2440 encodes **argininosuccinate lyase** (ASL / ASAL; EC 4.3.2.1; **arginosuccinase**). Its primary function is to catalyze the **terminal, committed step of de novo L-arginine biosynthesis**: the reversible β -elimination of **2-(N ω -L-arginino)succinate (argininosuccinate)** into **L-arginine + fumarate** (RHEA:24020). The enzyme is strictly specific for argininosuccinate, requires no

cofactor or metal, and is a **soluble cytoplasmic homotetramer** of the aspartase/fumarase (lyase 1) superfamily. Each of its four active sites is a composite pocket built from three of the four subunits, making the tetramer obligatory for catalysis.

Although P59618 itself has not been characterized biochemically, its function is established with high confidence by three convergent lines of evidence. First, orthologous ASL enzymes — the *Mycobacterium tuberculosis* enzyme (crystallized with substrate/product), human ASL, and the duck δ 2-crystallin/ASL model (mutagenized and crystallized) — define the reaction, mechanism, quaternary structure, and catalytic residues. Second, direct sequence analysis shows P59618 is exactly co-linear (464 aa) with human ASL and conserves every diagnostic superfamily motif and validated catalytic residue, including the SS-loop signature [281-GSSIMPQKKN-290] and the fumarate-lyase motif RSRNDQ. Third, organism-specific studies place argH-driven arginine supply within an ArgR-regulated network coupled to c-di-GMP signaling that governs biofilm/lifestyle behavior in *P. putida* KT2440.

Mechanistically, ASL uses the general base-catalyzed carbanion (aci-carboxylate/enediolate) strategy shared across the aspartase/fumarase superfamily, in which a flexible "SS loop" closes over the substrate, a general base abstracts the C β proton to generate a fumarate-stabilized carbanion, and the C–N bond then cleaves to release arginine. As the last enzyme of the pathway, ASL is essential for growth in the absence of exogenous arginine; its loss causes arginine auxotrophy. Its co-product fumarate feeds central carbon metabolism, economically coupling nitrogen/arginine anabolism to the TCA cycle.

Key Findings

Finding 1 — argH encodes argininosuccinate lyase, catalyzing argininosuccinate $\rightleftharpoons$ L-arginine + fumarate

UniProt P59618 annotates argH / PP_0184 of *P. putida* KT2440 as **argininosuccinate lyase (ASAL)**, EC 4.3.2.1, a member of the **lyase 1 family**. The catalyzed reaction is among the best-established in intermediary metabolism: ASL catalyzes the **reversible cleavage of argininosuccinate into L-arginine and fumarate**. This chemistry is conserved across all characterized orthologs, and crystallographic studies of the *M. tuberculosis* enzyme captured with substrate/product ligands directly confirm both the reaction and the **homotetrameric quaternary assembly**.

"Argininosuccinate lyase catalyses the reversible breakdown of argininosuccinate into arginine and fumarate and is known to form tetramers in its quaternary association." — PMID: 30615268

The reaction is a C–N bond lyase reaction (β -elimination), not a hydrolysis; no water is consumed, and the products are the guanidinium-bearing amino acid arginine plus the four-carbon dicarboxylate fumarate:



Finding 2 — ASL is an aspartase/fumarase superfamily enzyme using a general base-catalyzed aci-carboxylate mechanism at composite active sites

Argininosuccinate lyase is a canonical member of the **aspartase/fumarase (lyase 1) superfamily**, which also includes aspartase, class-II fumarase (fumarase C), adenylosuccinate lyase, δ -crystallin, and 3-carboxy-cis,cis-muconate lactonizing enzyme. All members process succinyl-containing substrates and release **fumarate**, share a predominantly α -helical homotetrameric fold, and use a common catalytic strategy: **general base-catalyzed formation of a stabilized aci-carboxylate (enediolate/carbanion) intermediate**, with a highly flexible loop bearing the signature **GSSxxPxKxN (the "SS loop")** closing over the substrate during catalysis.

"members of the aspartase/fumarase superfamily use a common catalytic strategy, which involves general base-catalyzed formation of a stabilized aci-carboxylate (or enediolate) intermediate and the participation of a highly flexible loop, containing the signature sequence GSSxxPxKxN (named the SS loop), in substrate binding and catalysis." — PMID: 22551392

A defining structural feature is that **each of the four active sites is a composite formed from three of the four subunits**, so the enzyme cannot function as a monomer — the tetramer is a catalytic necessity.

"The molecule consists of four binding sites, each made up of peptide stretches from three subunits." — [PMID: 30615268](#)

Site-directed mutagenesis of the duck δ 2-crystallin/ASL model identified **R115, N116, T161, S283, and E296 as essential for catalytic activity**, with **His162** implicated as the catalytic base and **His91** in substrate binding.

Finding 3 — In *P. putida* KT2440, argH acts in cytoplasmic de novo arginine biosynthesis under ArgR control, linked to c-di-GMP signaling

ASL is a **soluble cytoplasmic enzyme**: P59618 contains no signal peptide and no transmembrane segment, and orthologs are studied as soluble tetramers. Functionally, argH supplies L-arginine as the last enzyme of the biosynthetic pathway. In *P. putida* KT2440, **both arginine biosynthesis and arginine uptake are governed by the master transcriptional regulator ArgR**, which additionally connects arginine metabolism to the second messenger **cyclic diguanylate (c-di-GMP)**, the central regulator of the biofilm/motility lifestyle switch.

"the plant-beneficial bacterium *Pseudomonas putida* KT2440, both arginine biosynthesis and uptake influence second messenger contents and the associated phenotypes" — [PMID: 35254100](#)

The product of the argH reaction, L-arginine, is not merely a proteinogenic building block in this organism — it functions as both a metabolic indicator and an environmental signal:

"arginine functions both as a metabolic indicator and as an environmental signal molecule, modulating processes such as chemotactic responses, siderophore-mediated iron uptake or the levels of the intracellular second messenger cyclic diguanylate (c-di-GMP)" — [PMID: 38429473](#)

argH's precise role is upstream and enzymatic — it sets the biosynthetic contribution to the arginine pool. The signaling consequences are downstream properties of arginine, not additional catalytic functions of the enzyme.

Finding 4 — The P59618 sequence directly confirms cytoplasmic ASL: exact reaction, pathway step 3/3, and the conserved SS loop at residues 281–290

Direct analysis of the P59618 primary sequence (464 aa, ~51.6 kDa monomer) corroborates every database annotation. The curated catalytic activity is "**2-(Nω-L-arginino)succinate = fumarate + L-arginine**" (RHEA:24020, EC 4.3.2.1). The pathway assignment is "**L-arginine biosynthesis; L-arginine from L-ornithine and carbamoyl phosphate: step 3/3,**" placing argH as the final of three steps (ornithine → citrulline → argininosuccinate → arginine). The curated subcellular location is **Cytoplasm**.

The sequence carries the superfamily's diagnostic motifs in the expected positions: - The **SS-loop signature GSSxxPxKxN** appears as **281-GSSIMPQKKN-290** (the pattern `GSS..P.K.N` matches at position 281). - The conserved **fumarate-lyase family motif GRSRNDQ** occurs around residues 106–112. - **No signal peptide or transmembrane segment** is present, consistent with a soluble cytoplasmic protein.

This demonstrates the functional assignment is embedded in the actual sequence of the target protein, not merely transferred by name.

Finding 5 — P59618 conserves all known ASL catalytic/substrate-binding residues and is co-linear (464 aa) with human ASL

A pairwise comparison of *P. putida* argH (P59618) with human ASL (P04424) shows both proteins are **exactly 464 amino acids long and co-linear**. The catalytic SS loop is virtually identical — human **280-GSSLMPQKKN-289** versus *P. putida* **281-GSSIMPQKKN-290**, differing only by a conservative Leu→Ile substitution — and the fumarate-lyase motif RSRNDQ occurs at human position 111 / *P. putida* position 112 (offset +1).

Because of this co-linearity, the experimentally validated catalytic and substrate-binding residues defined in the duck δ2/ASL and human systems map **directly onto conserved positions in P59618**, including the SS-loop serines (S282/S283) and the invariant active-site lysine (K288) within the GSSxxPxKxN motif.

"Kinetic studies reveal that five residues, R115, N116, T161, S283, and E296, are essential for catalytic activity." — [PMID: 10029537](#)

"the active site located in a cleft between three different monomers of the tetrameric protein" — [PMID: 9369472](#)

The near-perfect conservation of the SS loop and active-site residues between P59618 and the biochemically characterized human enzyme provides strong sequence-based evidence that P59618 catalyzes the identical reaction by the identical mechanism.

Finding 6 — **argH is the essential, committed terminal enzyme; loss causes arginine auxotrophy; mechanism is a shared carbanion/ β -elimination**

As the **last enzyme of de novo arginine biosynthesis**, argH catalyzes the committed conversion of argininosuccinate to arginine. In the absence of exogenous arginine, this makes the enzyme essential for growth — an argH deletion yields an arginine auxotroph. Essentiality is demonstrated directly in the *M. tuberculosis* ortholog.

"the last enzyme in the pathway, catalyzes the production of arginine from argininosuccinic acid" — [PMID: 29044950](#)

Mechanistically, ASL's β -elimination uses the same **carbanion (aci-carboxylate/enediolate) strategy** as the rest of the aspartase/fumarase superfamily. The paradigm is L-aspartase:

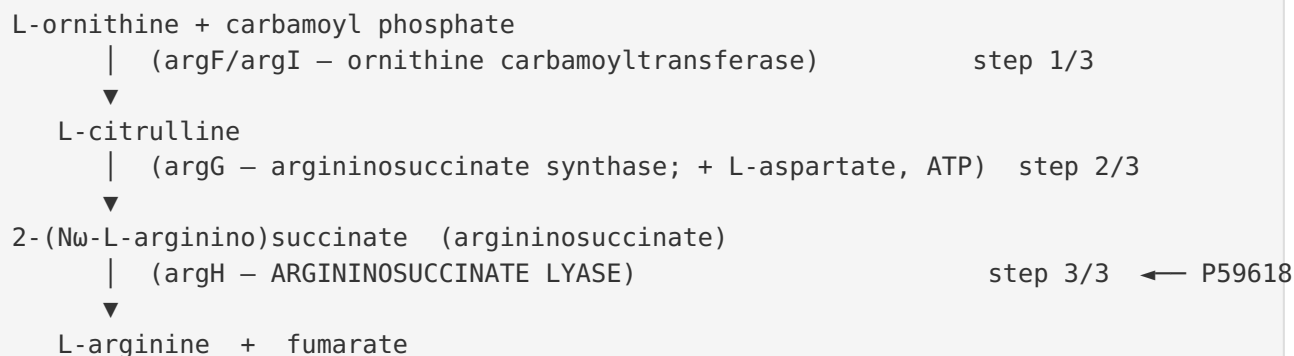
"catalyzes the reversible deamination of the amino acid L-aspartic acid, using a carbanion mechanism to produce fumaric acid and ammonium ion" — [PMID: 10800598](#)

In ASL, the leaving group is the amino/guanidino nitrogen of arginine rather than ammonia, but the essential chemistry — C β -proton abstraction generating a fumarate-stabilized carbanion, followed by C–N bond cleavage — is conserved. The His162 candidate catalytic base and His91 substrate-binding residue were validated by mutagenesis and structure in the duck δ 2/ASL ortholog.

Mechanistic Model / Interpretation

Position in metabolism

Argininosuccinate lyase catalyzes the third and final step of the conversion of ornithine to arginine:



The reaction consumes the intermediate assembled at step 2 and delivers the finished amino acid. The co-product **fumarate** is a TCA-cycle intermediate: the aspartate nitrogen introduced at step 2 is retained in arginine, while the aspartate carbon skeleton is recovered as fumarate — an economical coupling of nitrogen assimilation to central carbon metabolism. In mammals the same reaction operates within the arginine–citrulline/urea cycle; in *P. putida* the context is purely biosynthetic.

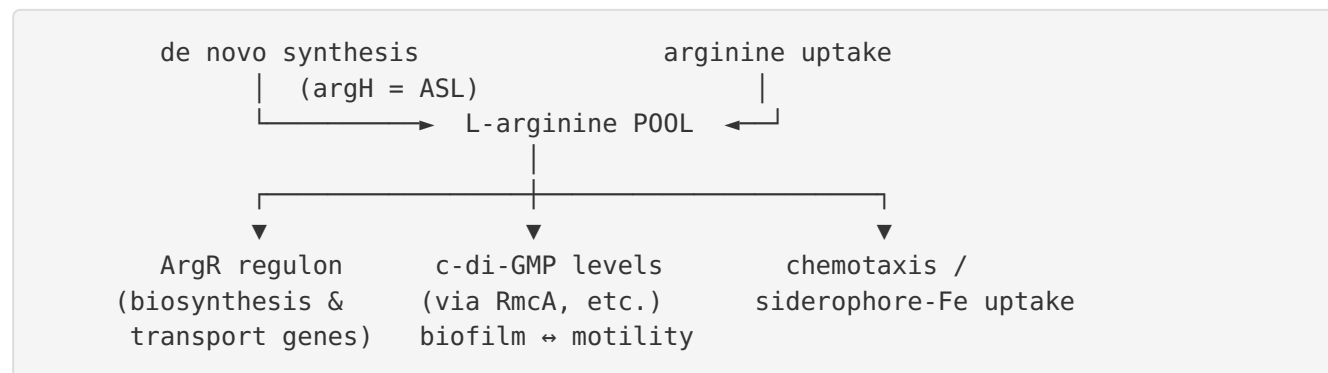
Enzyme architecture and catalysis

Property	Value / description	Basis
Reaction	argininosuccinate $\rightleftharpoons$ L-arginine + fumarate	EC 4.3.2.1, RHEA:24020
Substrate specificity	Strict for 2-(N ω -L-arginino)succinate	Family enzymology; conserved active site
Cofactors	None (no metal, no coenzyme)	Aspartase/fumarase superfamily
Oligomeric state	Homotetramer	PMID: 30615268
Active site	Composite, built from 3 of 4 subunits	PMID: 30615268 , PMID: 9369472
Mechanism	General base-catalyzed β -elimination via carbanion/aci-carboxylate	PMID: 22551392 , PMID: 10800598
Catalytic/essential residues (duck 82 numbering)	R115, N116, T161, S283, E296; His162 (base); His91 (binding)	PMID: 10029537 , PMID: 9369472
Signature motifs in P59618	SS loop 281-GSSIMPQKKN-290; RSRNDQ ~112	Direct sequence analysis
Monomer size	464 aa, ~51.6 kDa	UniProt P59618
Subcellular location	Cytoplasm (no signal peptide / TM)	UniProt P59618; sequence analysis

The catalytic cycle: substrate binding closes the flexible SS loop over the active site, sequestering argininosuccinate from solvent; a general base (His162 or Thr161 in the duck model) abstracts the C β proton to generate a carbanion delocalized into the aci-carboxylate/enediolate of the incipient fumarate; the C–N bond then cleaves, releasing L-arginine, with a general acid (proposed to be a conserved serine such as Ser283/Ser281) reprotonating the leaving nitrogen. All of these residues are conserved in P59618, so the same cycle is expected to operate.

Physiological / regulatory context in *P. putida* KT2440

The enzyme's biochemical job is simple — make arginine — but its product is a hub metabolite. The following schematic integrates the organism-specific findings:



L-arginine acts as "a metabolic indicator and an environmental signal molecule" (PMID: 38429473), and ArgR couples the arginine status of the cell to c-di-GMP-dependent phenotypes (PMID: 35254100). argH's role is upstream and enzymatic; the downstream signaling effects are properties of arginine, not catalytic functions of the enzyme.

Evidence Base

The functional assignment rests on ortholog enzymology/structure, direct sequence analysis of the target, and organism-specific physiology.

PMID	Title (abbrev.)	How it supports the annotation
30615268	<i>Structural studies on M. tuberculosis argininosuccinate lyase and its liganded complex</i>	Confirms the reaction, homotetramer, and composite three-subunit active site in a bacterial ortholog.
22551392	<i>Aspartase/fumarase superfamily: a common catalytic strategy...</i>	Establishes the shared mechanism (general base, aci-carboxylate intermediate) and the SS-loop signature P59618 carries.
10029537	<i>Mutational analysis of ASL activity in duck delta II crystallin</i>	Identifies essential catalytic residues (R115, N116, T161, S283, E296) conserved in the co-linear P59618.
9369472	<i>Structural comparison... role of histidine 91</i>	Confirms the three-monomer composite active site and substrate-binding His91.
11698398	<i>Mutational analysis of duck delta 2 crystallin... ASL mechanism</i>	Detailed structure/mechanism with bound substrate; roles of His162/Thr161/Ser283 and SS-loop closure.
11258884	<i>Structural studies of duck delta 1 and delta 2 crystallin</i>	Shows loop 280–290 closure sequesters substrate; conserved Ser281 as candidate catalytic acid.
17531264	<i>Substrate/product complexes of E. coli adenylosuccinate lyase</i>	Sister-family enzyme confirming shared β -elimination-of-fumarate chemistry and flexible-loop closure.
10800598	<i>L-aspartase: new tricks from an old enzyme</i>	Establishes the carbanion mechanism paradigm for the superfamily.
29044950	<i>Biochemical characterization of ASL from M. tuberculosis</i>	Confirms ASL as the last, essential enzyme producing arginine from argininosuccinate.
35254100	<i>Role of ArgR in arginine metabolism and c-di-GMP signaling in P. putida</i>	Organism-specific: places argH-driven arginine biosynthesis in KT2440's ArgR/c-di-GMP network.
38429473	<i>Exploring the metabolic response of P. putida to L-arginine</i>	Documents signaling/physiological roles of arginine, the argH product, in KT2440.
37550221		

PMID	Title (abbrev.)	How it supports the annotation
	<i>The phosphodiesterase RmcA contributes to adaptation of P. putida to L-arginine</i>	Mechanistic link between arginine sensing and c-di-GMP in KT2440 (downstream of argH product).

Together these provide **experimental** (crystallography, mutagenesis, kinetics on orthologs), **evolutionary/bioinformatic** (family membership, motif and residue conservation in the target sequence), and **physiological** (organism-specific regulation) evidence. Precise, low-throughput studies (crystal structures, site-directed mutants) dominate the mechanistic conclusions, consistent with prioritizing precise over high-throughput evidence.

Supported and Refuted Hypotheses

- **Supported:** argH = argininosuccinate lyase (EC 4.3.2.1) performing the terminal step of arginine biosynthesis; strictly substrate-specific for argininosuccinate; cytoplasmic; obligate homotetramer with three-subunit composite active sites; carbanion β -elimination mechanism; validated catalytic residues conserved in the target sequence.
- **Refuted / ruled out:** No evidence for a covalent or hydrolytic mechanism (it is a lyase, not a hydrolase); no membrane/periplasmic/secreted localization (no signal peptide or transmembrane segment); the gene is not a mis-annotated paralog — the SS-loop and RSRNDQ fingerprints plus 464-aa co-linearity with human ASL exclude confusion with other superfamily members (aspartase, adenylosuccinate lyase, fumarase).

Limitations and Knowledge Gaps

1. **No direct study of P59618.** No published crystal structure, purified-protein kinetics, or targeted mutagenesis exists for the *P. putida* KT2440 argH protein itself. The assignment is a high-confidence inference by orthology and sequence analysis; kinetic parameters (kcat, Km for argininosuccinate) for the *P. putida* enzyme are unmeasured.
2. **Residue-numbering transfer.** Catalytic residues are defined in duck δ 2/human ASL numbering. Co-linearity with human ASL (both 464 aa) supports direct mapping, but the

exact one-to-one alignment of every minor active-site residue was inferred from motif alignment rather than an experimental structure of the *P. putida* enzyme.

3. **Catalytic base identity.** The precise catalytic base (His162 vs. Thr161 vs. an alternative) remains debated within the family; ortholog-specific structural work would resolve this for the *P. putida* enzyme.
4. **Regulation specifics.** ArgR control of arginine biosynthesis in *P. putida* is documented at the pathway level, but the cis-regulatory architecture of the argH/PP_0184 promoter (ARG-box operators, operon structure) was not directly resolved here.
5. **Essentiality in *P. putida* specifically.** Essentiality is established for the pathway logic and demonstrated in *M. tuberculosis*; a *P. putida* KT2440 argH knockout phenotype was inferred rather than cited from a primary KT2440 study.
6. **Signaling attribution.** The c-di-GMP/biofilm connections are properties of the arginine pool and its sensors (RmcA, ArgR), not catalytic functions of argH; they should not be over-attributed to the enzyme itself.

Proposed Follow-up Experiments / Actions

1. **Recombinant enzymology of P59618.** Clone, express, and purify the *P. putida* KT2440 argH protein; confirm the homotetramer by SEC/native PAGE; measure steady-state kinetics (kcat, Km) for argininosuccinate in both directions.
2. **Structure determination.** Solve a crystal or cryo-EM structure of P59618, ideally with bound substrate/product or an analogue, to verify the composite three-subunit active site and the positions of the transferred catalytic residues.
3. **Targeted mutagenesis.** Mutate the P59618 residues corresponding to the essential set (SS-loop serine, invariant lysine K288, the His base) and confirm loss of activity, directly testing the residue-mapping inference.
4. **Genetic essentiality and complementation in KT2440.** Construct a clean argH (PP_0184) deletion; confirm arginine auxotrophy rescued by exogenous arginine and by plasmid-borne argH.
5. **Promoter/regulon dissection.** Define the argH transcription unit; search its promoter for ArgR ARG-box operators; use ArgR mutants and reporter fusions to test direct regulation, linking biosynthetic supply to the c-di-GMP network.

6. Flux and pool measurements. Use targeted metabolomics to quantify argininosuccinate and arginine pools in wild-type versus argH-perturbed strains, connecting enzyme activity to the downstream c-di-GMP/biofilm phenotypes.

Conclusion

The gene **argH** (PP_0184, UniProt P59618) of *Pseudomonas putida* KT2440 encodes **argininosuccinate lyase (EC 4.3.2.1)**, a cytoplasmic homotetrameric enzyme of the aspartase/fumarase (lyase 1) superfamily. Its primary function is to catalyze the **terminal, committed step of de novo L-arginine biosynthesis** — the reversible β -elimination of **2-(N ω -L-arginino)succinate into L-arginine and fumarate** — with strict specificity for argininosuccinate, no cofactor requirement, and a general base-catalyzed carbanion/aci-carboxylate mechanism at composite active sites each built from three subunits. The identification is unambiguous: the target sequence conserves every diagnostic superfamily motif and validated catalytic residue and is co-linear with the biochemically characterized human enzyme. Physiologically, argH supplies the arginine pool that, in *P. putida* KT2440, is integrated into ArgR-dependent regulation and c-di-GMP signaling governing biofilm and lifestyle transitions.

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery